

2 (a) (i) $p \propto T$ or $pV/T = \text{constant}$ or $pV = nRT$

C1

$$T (= 5 \times 300 =) 1500 \text{ K}$$

A1 [2]

(ii) $pV = nRT$

$$1.0 \times 10^5 \times 4.0 \times 10^{-4} = n \times 8.31 \times 300$$

or

$$5.0 \times 10^5 \times 4.0 \times 10^{-4} = n \times 8.31 \times 1500$$

C1

$$n = 0.016 \text{ mol}$$

A1 [2]

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- (b) (i) 1. heating/thermal energy supplied B1
2. work done on/to system B1 [2]
- (ii) 1. 240 J A1
2. same value as given in 1. (= 240 J) **and** zero given for 3. A1
3. zero A1 [3]